5

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2

2

1

1

D D

C C

B B

A A

兼容DCV1117

2N7002 TS4 TS4+ B24 TS4+ TS1+

VC24 VC23 VC1 VC2 VC3 VC4 VC5 VC6 VC7 VC8 VC9 VC10 VC11 VC12 VC13 VC14 VC15 VC16 VC17 VC18 VC19 VC20 VC21 VC22 VC0

LDSG

LCHG

AFE_SDA

AFE_SCL

LOAD_Activation B24 AFE_SCL MCU_SCL AFE_SDA MCU_SDA
TS1 TS2 TS3 TS4 TS2 TS1 TS3 TS1+ TS2+ TS3+ TS2+ TS3+ SRN

SRP

LOAD_Activation B13 B14 VC0 VC14 VC13 VC12 VC15 VC8 VC7 B5B7 B5 VC6 B6 B9B8 B2 B10 B15 B13 VC4 B14 B11 B4 B12 B3 VC3 VC5 VC2 B1 VC1 B10 B0

VC10 B5B6B7B8B9 B11 B12 VC5 B13 B14B15 B10 VC9 VC10 VC11 B2B5B10 B1B4B12 B0B8B7B11 B9B3B6 B15 B24 B19 B23 B18 B22 B17 B21 GPS_VCC

B16 B20 B17 B18 B19 B20B21 B22 B23 B24 VC20 B17 B19 B20 B23 B21 B24 B22 VC23 VC22 VC21VC20 B20 VC19 VC18 VC17 VC16 B15 VC15 VC24 B15 B16 B18 B16 CAN_H

CAN_L DEG_RX DEG_TX

485_B_TX

485_A_RX

485_A_RX CAN_H

485_B_TX CAN_L CAN_H

485_A_RX COM_S CAN_L

485_B_TX

485_RE CAN_RX CAN_TX

485_A_RX

485_B_TX

TTL_RX

TTL_TX

SW_ON

SW+PRE_VOL PRE_VOL_EN B24

LOAD_ON

DIN

CHG_ON

CIN

ACC_ON

ACC

TRS

SW+ CIN

DIN
ACC

SW- KEY+ HT_EN

VIBRATE VIBRATE

TS6

LED1_EN LED1 LED2_EN LED2

LED3 LED4

LED5

LED3_EN MCU_SCL MCU_SDA TTL_RX TS5 TS6

LED5

LED4

LED3

LED2

LED1 GND BUZ_EN

CMF DMF

LED4_EN

LED3_EN
LED2_EN
LED1_EN

LED5_EN TRS_RX TRS_TX LOAD_ON COM_S PRE_VOL_EN PRE_EN SW_ON

KEY+

ACC_ON

TEM_EN

CHG_ON

BUZ_EN BUZ+ HT_EN

CHG_IN GND

NRSTA

SWCLK

SWDIO

TS5 TEM_EN

LED5_EN

LED4_EN MCU_SDA MCU_SCL CAN_RX CAN_TX

PRE_VOL DEG_RX DEG_TX

BL_EN

TTL_TX COM_EN GPRS_EN MCU_WAKE NRSTA SWCLK

SWDIO

LOAD COMV_EN 485_RE COMV_EN

COM_EN SRN SRP

CVS DVS
PRE_EN

TRS_BP TRS DVS CVS CMF DMF

VCC_IN MOS_VCC

VREG

VDD_MCUA

P- P- VCC_IN

VDD_MCUA

VCC_IN GPS_VCC

ISO_5V COM_3.3V ISO_5V

VDD_MCUA

EXT_VCC

VDD_MCUA

ISO_5V

ISO_5V COM_3.3V

ISO_5V

VDD_MCUA

EXT_VCC

EXT_3.3V

EXT_3.3V EXT_VCC

VDD_MCUA

VDD_MCUA

B- D_C

VDD_MCUA

VDD_MCUA

VDD_MCUA

VDD_MCUA

P- P- VDD_MCUA

VDD_MCUA

VDD_MCUA

VDD_MCUA B+ VDD_MCUA

P- B+ VDD_MCUA

LCD_12V

VDD_MCUA

VDD_MCUA

VDD_MCUA

VDD_MCUA

VDD_MCUA VDD_MCUA

VDD_MCUA COM_3.3V

VDD_MCUA

VDD_MCUA

VDD_MCUA

VDD_MCUA

VDD_MCUA

5V

5V GPS_VCC

12V

ISO_5V

12V

LCD_12V

5V MOS_VCC MOS_VCC

VREG

B- P- MOS_VCC

D_C

P- VDD_MCUA

VDD_MCUA

P- P- VDD_MCUA

VDD_MCUA

B24

LCHG

LDSG

B24 VC24 VC23 VC22 VC21 VC20 VC19 VC18 VC17 VC16 VC15 VC14 VC13 VC12 VC11 VC10 VC9 VC8 VC7 VC6 VC5 VC4 VC3 VC2 VC1 VC0 MCU_SCL MCU_SDA

SRP

SRN

KEY+ CHG_IN
VC14 VC13 VC12 VC11 VC10

B15 VC15 VC8 VC7VC6 VC5

VC5 VC4 VC3 VC2 VC1 VC0
VC10 VC9 VC20 B24 VC15 VC23 VC22 VC21 VC20 VC19 VC18 VC17 VC16 B15 VC24 DEG_RX DEG_TX COM_S

485_RE CAN_RX CAN_TX

TTL_TX

TTL_RX BL_3.3V/12V

LOAD_ON

SW_ON

LOAD

PRE_VOL_EN

PRE_VOL B24 KEY+

LOAD_ON

CHG_ON

ACC_ON

TRS MCU_WAKE HT_EN

VIBRATE VIBRATE MCU_SCL MCU_SDA

TTL_TX TTL_RX CMF DMF TRS_RX TRS_TX LOAD_ON COM_S PRE_VOL_EN PRE_EN SW_ON
KEY+

ACC_ON

CHG_ON HT_EN

CHG_IN

CAN_TX CAN_RX DEG_RX DEG_TX

BL_EN

COM_EN GPRS_EN MCU_WAKE COM_EN

PRE_VOL

LOAD 485_RE COMV_EN

BL_EN

BL_3.3V/12V

B24 GPRS_EN

BL_3.3V/12V COMV_EN

COM_EN

BL_3.3V/12V

SRN SRP CVS DVS
LDSG

LDSG

PRE_EN

LCHG

TRS_TX

TRS

TRS_RX DVS CVS CMFDMF

TP80 R468 R467 ZD68 K A R296

J13 1
2

3
4

5 ZD42 A K Q81 S

S1S2S3S4S5S6 G

D

R290 R306 Q303 R365

2

1

3 R801 R197

R176 R198

R174 Q40

3

1

2 R180 R182 R160 R167 R150

R108 R98 D21 C13 MASK4 R86 R55 R79 D11 Q18

1

2 3 R34 R37 Q17

1

2 3 R36 R406 R144 C51 R267 U7 RO

1 DI 4 GND

5 VCC

8 RE

2 DE

3 A
6 B

7 D700 TP49

U15 VBAT

1 PC13-TAMPER-RTC

2 PC14/OSC32_IN

3 PC15/OSC32_OUT

4 PF0-OSC_IN

5 PF1-OSC_OUT

6 NRST

7 PC0/IN6 I2C3_SCL

8 PC1/IN7 I2C3_SDA

9 PC2/IN8

10 PC3/IN9

11 VSSA1

12 VDDA1

13 PA0/IN1_WKUP

14 PA1/IN2/RX2

15 PA2/IN12/TX2

16 PA3/IN3/RX2 17 VSS2 18 VDD2 19 PA4/IN1/I2C2_SCL 20 PA5/IN5/I2C2_SDA 21 PA6/IN3 22 PA7/AN7 23 PC4/IN5/TX 24 PC5/IN12/RX 25 PB0/TIM3 26 PB1/IN3 27 PB2-BOOT1 28 PB10/SCL2/TX 29 PB11/SDA2/RX 30 VSS3 31 VDD3 32 VDDIO2

48 VSS4

47 PA13/SWDIO

46 PA12/CAN_TX

45 PA11/CAN_RX

44 PA10/RX1

43 PA9/_TX1

42 PA8

41 PC9/TIM8_CH4

40 PC8/TIM8_CH3

39 PC7/I2C4_SDA

38 PC6/I2C4_SCL

37 PB15/TIM1_CH3N

36 PB14/TIM1_CH2N

35 PB13/CAN2_TX

34 PB12/CAN2_RX

33 VDD5 64 VSS5 63 PB9/TIM4_CH4 62 PB8/TIM4_CH3 61 BOOT0 60 PB7/SDA1/RX1 59 PB6/SCL1/TX1 58 PB5/I2C1_SMBA 57 PB4/RX5 56 PB3/TX5 55 PD2/RX5 54 PC12/TX5 53 PC11/RX3/RX4 52 PC10/TX3/TX4 51 PA15/RX2 50 PA14/TX2/SWCLK 49 TP59

BT2 R216

Q60

2

1

3 Q62

3

1

2 C508 C504 C131 R381 R706 R459 Q91

3

1

2 D46 A K Q80 S

S1S2S3S4S5S6 G D

R350 R353 C67 ZD33 A K

ZD30 K A C56 D17

R124 R126 R104 R107 R6 C16 D15

1 2 C6 C15 MASK1 D2 Q22

1

2 3 Q19

1

2 3 R68 R53 Q20

1

2 3 R52 TP16 R408 C49

J104

1
2
3

4 R720 R717 TP62
C72 R214 R219 C77 R227 R252 R380 R504 R372 R509 R512 R374 C123 C130 R255 R333 D49 A K C112 RS15 RS3 Q75 S

S1S2S3S4S5S6 G D

R299 RS7 C115 Q520

2

1

3

D23 U12

1 2

4 3 Q35

1

2 3

TP29 R94 R31 C201 C47 TP79 R54 R56 R71 R44 R39 TP20

J2
1
2
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4
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14
15

R716 D306 A K D304 A K

C76 TP63

D67 ZD35 A K R217 R204

R258 Q501 1
2 3 R501 D94 A K Q502

3

1

2

R314 RS18 D51 A K C118 R301 D48 A K C117

R192

J19
1

2 C63 R151 U11 +

3

- 4 V- 2 V+

5 Vout 1 ZD29 A K

TP34 R119 C38 C35 Q29

1

2 3 R97 Q9

1

2 3 TP8 R2 R787 R325 ZD1 A K

J1
1
2
3
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7

8 Q503

1

2 3

R81 R35 R60 R83 TP23 D301 A K R145 C43 R141 R235 R225 C93 C165

L6 C503 D66 R334 D303 R293 C120 Q83

1

2 3 D44
1 J16
2

3
4

5RS20 Q85 S

S1S2S3S4S5S6 G

D

D55 A K

R195

J7
1
2

Q38

1

2 3 R179 Q42

3

1

2 R188 C62 R162 C53 R157

TP35 R115 R105 R121 R91 Q26

1

2 3 R103 D5 C1 U1C24

1 C23
2 C22
3 C21
4 C20
5 C19
6 C18
7 C17
8 C16
9 C15
10 C14
11 C13
12 C12
13 C11
14 C10
15 C9
16 C8
17 C7
18 C6
19 C5
20 C4
21 C3
22 C2
23 C1

24 C0

25 VSS

28 SRN

26 SRP

27 GP6/LDSG

29 GP5/LCHG

30 GP4/AN4

31 GP3/AN3

32 GP2/PREL/AN2

33 GP1/AN1

34 V1P8

35 V3P3

36 SDA

37 SCL

38 VREG

39 VBASE

40

LOAD

41 PDSG

42 PCHG

43 PACK

44 DSG

45 CHG

46 VCP

47 VTOP

48 R42 C27 C20C32 Q25

1

2 3 TP26

C50 R130 A

ZD82 K R712

R453 R212 C78

Q500

1

2 3 R329 U27 1 2

4 3

R465 R461 R287 RS17 RS6 C119 Q82 S

S1S2S3S4S5S6 G

D

R292 R457

1 Q301

2 3

D25 K1 ZD34 A K

ZD32 A K

TP36 Q32

1

2 3 R92 R27 D9 2 1 C3 C5 D126 R328 C17 TP27 C24 R38 R61 Q21

1

2 3 R78 R40 R127 TP42

U2 VDD1

1 VOA

2 VIB

3 GND1

4 VDD2

8 VIA

7 VOB

6 GND2

5 R715 R722 R721 R215 Q49

2

1

3

R206

D32 R253 R379 R259 D71 R511 U18 GND

1

IN

2 OUT

3 C124 D62 R373 C113 R318 D50 A K Q308

1

3 2 Q307

3

1

2

C66 R189 D22 R184 R159

R114 C36 C39 Q27

1

2 3 R116 R123 Q706

1

2 3 ZD2 A K D7 2 1 D129 R23 D131 R63 R45 C28 R41 R134 R139 R666 R709 R229

Y2 TP60

C75 C69

R254 CD1 R708 R469 R466 R315 D54 A K R308 R360 R303

R726 R190R199

R169 C61 C54 R164 R166

Q34 1

2 3 R110 R112 R785 R888 C203 R12 D130 R72 C26 R77 C29 C19 R84 R58 R85

TP17 R131 D302 A K U9

1 2

4 3 U3 D

1 GND

2 VCC

3 R

4 RS

8 CANH

7 CANL

6 Vref 5 U20

1 2

4 3

C82 TP58 TP56 R232 Q58

2

1

3 R256 C148 ZD500 A K C501 C505 C142 R506 R458 Q76 S

S1S2S3S4S5S6 G

D

R300 D52 A K

1 Q78

2 3 R364 R362

R200 Q44

2

1

3 R178 C60 R152

TP31 R117 C34 C41 R100 Q30

1

2 3

C12 D12

1 2 R67 R70 TP21

U5 VDD1

1 VOA

2 VIB

3 GND1

4 VDD2

8 VIA

7 VOB

6 GND2

5 R140 R142 R143 R265

ZD81 A K
TP65

C70 R213

C125 R464 RS16 Q88

1

2 3

RS4 R289 RS8 D300 Q302

3

1

2 R356

R187 Q41

3

1

2

R156

TP33 R106

J32

1
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4
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9
10

11 Q39

1

2 3 C14 R339 MASK2 ZD55 A K

ZD50 A K C22 R69 C33 R57 TP24

TP40

C45 R724 R714 R718 R221 C73 R203 ZD36 A K

R230 Q48

3

1

2 R502 C126 R326 Q143

1

2 3 Q800

1

2 3

R320 R304 RS2 Q87 S

S1S2S3S4S5S6 G

D

R298 D45 Q45 1

2 3

J6

1
2
3
4
5
6

7 C59 R725 R181 R161 R163 C52

TP38 R109 R95 R3 D127 R600 ZD49 A K C7 R28 R73 R64 R88 Q24

1

2 3 Q15

1

2 3 R47 Q11

1

2 3 R43 TP22 R129 R719

D64 Q47

2

1

3 R207 C79

D500 R513 R508 RS19 Q77 S

S1S2S3S4S5S6 G

D

R456 R309

Q46

1

2 3

TP53 F1D18 C58 C65 R185 R165 R154

TP39 R99 C37 R111 R96 Q31

1

2 3 Q28

1

2 3

MASK3 C18 Q12

1

2 3 R82 C23 ZD17 A K R89 R46 Q14

1

2 3 R409 R407 TP43

C46 D307 A K R346

TP50

J5

1

2

3

4 C80 TP64

J9

1

2 R218 C81

J8
1
2
3
4

5 C71

D30

1 2 R260 U41 FB

1 NC

2 VIN

3 BST

4 SW

5 DIM

6 EN

7 GND1
8 GND2

9 C149 R500C94 R503 C164 R707 C92 R460 Q89

2

1

3 Q84 S

S1S2S3S4S5S6 G D

R291 C116 R351

R196 ZD31 A K

U13

1 2

4 3 R153 Q36

1

2 3 R113 C42 Q33

1

2 3
R30 D10 2 1 D705 R13 R75 R59 R51 R66 TP15

C48 R135 R222 TP61 R257 R705 D41 A K

TP78 Q92

1

2 3 D37 C114 R294 QF1

2

1

3 ZD41 A K R311 RS5 Q86 S

S1S2S3S4S5S6 G D

R355 R354

1 J12
2

3
4

5 D132

R201 R186 R183 R158

R120 C40 R93 R90 R463 ZD52 A K D706 C21 R48 R87 R74 Q23

1

2 3

TP19

R133 U10

1 2

4 3 R730

R220 R234
TP57 Q59

2

1

3 U37 VIN1

9 VIN2

10 EN

8 GND1

1 VOUT1
5 GND4

4 GND5

3 GND6

2 GND2

11 GND3

12 VOUT2

6 VSEL

7 R321 R316 RS1 Q79 S

S1S2S3S4S5S6 G D

R302 R800 R194

R168 D16 C55 R155

TP30 R101 C11 R29 D707 C2 R601 C200 C10 D8 2 1 ZD20 A K
R49 R76 C31 C30 TP18 C300 U6 1 2

4 3 C44R136 R710 R711

J10

1
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7
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9 TP66 R233 R269 Q61

3

1

2 R510 R507 C128 R251 Q700

1

2 3 ZD67 K A D26 R305 R307 R312

D24 Q43

1

2 3

TP32 R125 R474 R786 R472 C820 R62 R50 C25 R65 R80 TP25

R137 TP41 D61 Q702

2

1

3 D305 A K

ZD83 A K

R224 Y1 R205

R371 R382 Q304

3

1

2 R288 R310 Q74 S

S1S2S3S4S5S6 G D

R297 R352

J17 1
2

3
4

5 C800 R193 R191 C64

TP37 R122 R118 R102 R32 Q10

1

2 3 R335 R470 C8 C9 R471 R327 Q3

1

3 2

Q16
1

2 3 Q13

1

2 3

TP14

R713 Q703

3

1

2 R223 R454 ZD37 A K R226

C74

D34 D72 R275 R278 C500 D31 R261